



NEZHA BAHKOUK

Electronics Engineer specializing in AI

Phone

+212696333442

Email

nezhabakhkouk.19@gmail.com

Address

Agadir, Morocco

About Me

Specialized in machine learning, holding a master's degree in Control and Industrial Computing (Electronics and Telecommunications). Passionate about modern technologies and their applications in manufacturing environments. Recently completed a six-month internship at Aptiv in Tangier, within the Cutting & LP department, where I participated in a continuous improvement project using AI-based dynamic speed control to enhance production efficiency.

Languages

- Tamazight: Native
- Arabic: Fluent
- French: advanced
- English: advanced
- Russian: Beginner

Awards

Lean 2.0 OPEX Tournament – Local Final Winner Aptiv M6, Tangier, Morocco | May 2025

- Awarded 1st Place for leading the project "Dynamic Speed Control of Komax Cutting Machines Using Artificial Intelligence."

Experience

AI/ML Engineer Intern – Aptiv Company, Tangier, Morocco Feb 2025 – July 2025

- Developed a dynamic speed control system for cutting machines using AI techniques to improve industrial performance. Analyzed operational data and contributed to the Lean OPEX 2.0 continuous improvement project.

🏆 1st Place Winner – Aptiv Lean OPEX 2.0 Tournament, Tangier (May 2025)
Project: Dynamic Speed Control of Komax Cutting Machines Using AI

Field Researcher – High Commission for Planning (General Census of Population and Housing)

Sept 1 – Sept 30, 2024

- Contributed to field and organizational tasks as part of a team.
- Enhanced teamwork, discipline, and professional commitment.

Academic Project – Cadi Ayyad University, Marrakesh Feb 2023 – July 2023

- Antenna Optimization:
- Improvement of reflection performance using metaheuristic algorithms, with simulations in HFSS and MATLAB

Academic Project – Ibn Zohr University, Ait Melloul

Feb 2021 – June 2021

- Developed a word prediction model based on Recurrent Neural Networks (RNN).
- Used PyTorch and Scikit-learn to improve linguistic performance and data analysis.

Professeure et Assistante du Surveillant Général – Collège Sidi Khalil, Chtouka Ait Baha

Feb 2021 – June 2021

- Managing and organizing the student dormitory department
- Contributing to the implementation of rules and discipline within the institution.
- Supervising daily activities and coordinating between students and administration to ensure a safe and well-organized educational environment.

Hard skills

Artificial Intelligence & Machine Learning:

- Linear / Logistic Regression, Random Forest, SVM, PCA
- Deep Learning (ANN, RNN, LSTM)
- Python, PyTorch, Scikit-learn, Data Processing & Statistical Analysis

Electronics & Electrical Engineering:

- Signal Processing, Communication Systems
- Antenna Design and Optimization
- Simulation with HFSS, CST, MATLAB
- RF & Microwave Techniques
- Electric Machines, Digital and Power Electronics

Software & Tools:

- MATLAB, HFSS, CST, ...
- PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Jupyter, Python, ...
- SQL, LaTeX, Word, PowerPoint, Excel, Canva...

Soft skills

- Leadership
- Teamwork
- Professional Communication
- Creativity
- Patience and Stress Management
- Analytical Thinking
- Problem Solving
- Critical Thinking
- Curiosity

Education

PhD Candidate – Laboratory of Intelligent and Sustainable Technologies (LaRTID)

2023 – Present

National School of Applied Sciences, Cadi Ayyad University – Marrakesh

Master's Degree – Industrial Control and Computing (Electronics and Telecommunications)

2021 – 2023

National School of Applied Sciences, Cadi Ayyad University – Marrakesh

Bachelor's Degree – Physical Sciences (Electronics)

2018 – 2021

Faculty of Applied Sciences Ait Melloul, Ibn Zohr University

High School Diploma – Physical Sciences

2015 – 2018

AL Quds High School, Chtouka Ait Baha

Training Courses & Certificates

🏆 1st Prize at the local Lean 2.0 OPEX Tournament at Aptiv M6, Tangier, Morocco, for the project: Dynamic Speed Control of Komax Cutting Machines Using Artificial Intelligence | May 2025

- Online Training – AI Hababi Business for E-Services & Entrepreneurship (UK)
Course: Online Freelance Strategy & AI | November 2024
- 🎓 Coursera Certificate – IBM
Course: Introduction to Artificial Intelligence | March 7, 2024
- 🎓 Participation Certificate – ICESCO
• Participation in the 3rd edition of the Master Class on Scientific Research, organized by the ICESCO Chair for Open Education – École Normale Supérieure, Marrakech | June 4–6, 2024
- 🎓 Certificate – Queen Mary University of London
Course: Research Methodology | March 16, 2024
- 🎓 Participation Certificate – Clarivate, UAE
Course: Artificial Intelligence in Web of Science and Creation of Scientific Research Profiles | May 30, 2024
- 🎓 Certificate – University of Virginia (USA) in collaboration with BCG
Course: Digital Transformation | March 16, 2024
- 🎓 Certificate in Education and Pedagogy – LCP Laboratory in collaboration with the Gueliz Institute for Educational Specialties | May 2022
- 🎓 Training Program – International Center for Leadership in Human Development, Tangier, Morocco
Topics: Keys to Success and How to Influence Others | June 2020
- 🎓 Online Training – High Commission for Planning (Morocco)
Topic: Population and Housing Statistics | March – June 2024